

Assessment 2 Individual Workbook

Section 1: Personal contributions to the team design portfolio

User story 1:

Stub: Order Fruit Groups

User Story:

As a kindergarten student, I want to arrange the fruit groups from 1 to 5 so that I can improve my counting and ordering skills.

Acceptance Criteria

1. The game shows five fruit groups with different quantities.
2. The student can drag and drop the fruit groups into the correct order from 1 to 5.
3. The game gives feedback after the student completes the order.

Story Size: S

URL to prototype (Figma prototype design):

<https://www.figma.com/board/wcrZx8euXC17OJRNfkbkiT/2026-TR2-ICT1-Group-1?node-id=1-2&p=f&t=XWw5kRtiGiBaMrmO-0>

URL to the game:

<https://www.kopo.com/g/game.php?gno=14378&qid=8652&lid=5373&sid=2965>

Sprint Pilot Backlog

ID	As a	I want to able to play	So that
1	Kindergart en student	Arrange the fruit groups from 1 to 5.	My ordering and counting skills can improve
2	Kindergart en student	Drag each fruit group into the correct position	I can understand the order of different quantities
3	Kindergart en student	Receive feedback after completing the order	I can know whether my arrangement is correct
4	Kindergart en student	Try the game again after a wrong arrangement	Mistakes can help me learn

Feedback:

1. Improve question. It should be about ordering? "Order By numbers in Order"
2. Improve the images more fun and interesting
3. You can also add animation to make it more interesting

User story 2:

Stub:

Order Animals from Smallest to Biggest

User Story:

As a kindergarten student, I want to arrange animals from smallest to biggest so that I can understand size comparison and ordering.

Acceptance Criteria:

1. The game displays animal cards in a mixed order.
2. The student can drag the animal cards into empty spaces.
3. The animals must be ordered from smallest to biggest.

Story Size: M

URL to prototype (Figma prototype design):

<https://www.figma.com/board/wcrZx8euXC17OJRNfkbkiT/2026-TR2-ICT1-Group-1?node-id=1-2&p=f&t=XWw5kRtiGiBaMrmO-0>

URL to the game:

<https://www.kopo.com/g/game.php?gno=14378&qid=8665&lid=5373&sid=2965>

Sprint Pilot Backlog

ID	As a	I want to able to play	So that
1	Kindergart en student	Arrange animals from smallest to biggest	Size comparison becomes easier to understand
2	Kindergart en student	Drag animal cards into empty spaces	The activity can be completed independently
3	Kindergart en student	Compare the sizes of different animals	Small and large animals can be recognised correctly
4	Kindergart en student	Receive feedback after completing the order	Correct and incorrect arrangements can be identified

Feedback:

1. Because you are make prototypes for KOPO game, it should following the same template.
2. Improve the images more fun and interesting
3. You can also add animation to make it more interesting
4. You can make 3 ordering games instead

User story 3 :

Epic: Ordering Game

Stub:

Order Vehicles from Smallest to Biggest

User Story:

As a kindergarten student, I want to arrange vehicles from smallest to biggest so that I can improve size comparison and logical ordering skills.

Acceptance Criteria:

1. The game displays vehicle cards in a mixed order.
2. The student can drag the vehicle cards into empty spaces.
3. The vehicles must be ordered from smallest to biggest.

Story Size: M

URL to prototype (Figma prototype design):

<https://www.figma.com/board/wcrZx8euXC17OJRNfkbkiT/2026-TR2-ICT1-Group-1?node-id=1-2&p=f&t=XWw5kRtiGiBaMrmO-0>

URL to the game:

<https://www.kopo.com/g/game.php?gno=14378&qid=8666&lid=5373&sid=2965>

Sprint Pilot Backlog

ID	As a	I want to able to play	So that
1	Kindergart en student	Arrange vehicles from smallest to biggest	Size comparison becomes easier to understand

2	Kindergarten student	Drag vehicle cards into empty spaces	The activity can be completed independently
3	Kindergarten student	Compare the sizes of different vehicles	Small and large vehicles can be recognised correctly
4	Kindergarten student	Receive feedback after completing the order	Correct and incorrect arrangements can be identified

Feedback:

1. Add strokes to make the images more visible
2. Other looks ok
3. Add sounds if can

Section 2: Rationale Behind Key Design Decisions

2.1 Project Goals and Initial Concept

Following the client interview, the main **Project Goal** was to create educational games that were engaging, motivating, and suitable for **kindergarten students**. My individual **Deliverables** included three Figma prototypes, three functional KOPO games, User Stories, Acceptance Criteria, and a Sprint Pilot Backlog. Several possible game types were considered during the initial planning stage, but an **Ordering Game** was selected because one consistent interaction could support different learning outcomes without requiring young students to learn complicated controls. The concept was then divided into three themes: fruit quantities, animal sizes, and vehicle sizes. Fruit was selected for the first game because familiar objects could help children connect visible quantities with numbers from 1 to 5. Digital activities can support young children's understanding of numbers when the learning objective is clear and the interaction is appropriate for their age (Papadakis et al., 2018). The animal and vehicle games extended the same ordering structure to size comparison, creating variety while maintaining **Design Consistency**. Consistent and meaningful game mechanics can support learning engagement when they are connected to clear educational goals (Plass et al., 2015).

2.2 Developing the Game Requirements

The initial concepts were developed into clearer **User Stories** from the perspective of kindergarten students. Each story followed the structure "As a kindergarten student, I want to..., so that I can...", which connected the required action with a specific learning benefit. User Stories were useful because they kept each requirement brief while still supporting discussion, clarification, and later refinement (Cohn, 2004). The main stories were then divided into smaller actions, including arranging objects, dragging items into empty positions,

comparing quantities or sizes, receiving feedback, and trying again after an incorrect answer. These actions were converted into **Acceptance Criteria** so that each KOPO game could be checked against specific requirements instead of relying only on personal judgement. **Story Size** was also used for basic **Story Estimation**. The fruit game was rated S because the correct order could be identified directly by counting from 1 to 5, whereas the animal and vehicle games were rated M because suitable images with clear size differences had to be selected. Designing the requirements from the learner's perspective also supported **User-centred Design**, which aims to create systems that match the abilities and needs of intended users (International Organization for Standardization, 2019).

2.3 Prototyping and Iterative Improvement

Figma was first used for **Software Prototyping** to plan the layout, image positions, instructions, and empty drop areas before the games were developed in KOPO. Building the functional KOPO versions made it possible to compare the planned interfaces with the actual appearance and interaction of each game. This process supported **Iterative Refinement**, because design problems that were not obvious in the early prototype became clearer during implementation. Involving children's needs in technology design and reviewing how they may understand an interface can improve the suitability of the final product (Druin, 2002). Feedback then resulted in several specific improvements. The fruit question was rewritten to clearly instruct students to order the groups from 1 to 5, while the images were made more interesting and simple animation was considered. For the animal game, the same KOPO template was retained across all three activities, and the images were improved without changing the ordering mechanic. Stronger outlines were added to the vehicle images because some objects did not stand out clearly from the background. Immediate correct or incorrect responses and the opportunity to retry were also retained because clear formative feedback can help learners identify mistakes and improve their next attempt (Hattie & Timperley, 2007; Shute, 2008).

2.4 Final Design Decisions and Trade-offs

Several final decisions involved balancing visual interest, learning value, and technical limitations. One important **Design Trade-off** concerned the use of colourful images, animation, and sound. Carefully selected multimedia can support attention and interaction, but unnecessary visual or audio elements may distract learners from the main task (Mayer & Fiorella, 2021; Moreno & Mayer, 2007). Therefore, **Core Functionality** was prioritised, including clear instructions, visible drop areas, simple drag-and-drop interaction, immediate feedback, and the opportunity to retry. The same layout and control method were maintained across the three games to improve **Acceptability**, because a familiar interface would be easier for kindergarten students to understand and use. **Time Constraints** and **Platform Constraints** limited the use of complex animation and sound, so these features were treated as optional improvements instead of essential requirements. The final designs followed an **Agile Process**, as the games were gradually adjusted through feedback, review, and **Continuous Improvement** rather than remaining unchanged from the first concept (Schwaber & Sutherland, 2020).

Section 3: Reflection on Feedback, Iteration, Constraints and Design Trade-offs

3.1 Feedback and Iteration

Prototype reviews made me realise that a game could work technically but still be unclear to young learners. In the first fruit activity, more attention was given to the images and drag-and-drop function, while the instruction did not clearly state that the groups should be ordered from 1 to 5. After receiving this **Feedback**, the question was rewritten in a more direct way. A similar issue appeared in the animal prototype because its layout did not fully match the KOPO template used in the other games. Keeping the same structure across all three activities improved consistency. For the vehicle game, clearer outlines were added because several images blended into the background. These revisions demonstrated **Inspection**, **Adaptation**, and **Iteration**, since every change addressed an actual problem rather than adding unnecessary features. Regular review also prevented the first version from being treated as the final version (Schwaber & Sutherland, 2020).

3.2 Evaluation of Outcomes

Overall, most core **Learning Outcomes** were achieved. Each game included an ordering task, drag-and-drop interaction, feedback, and an opportunity to retry. Reusing the same layout also supported **Usability**, because students could focus on counting or size comparison instead of learning new controls in every activity. However, not every planned feature was completed. Animation remained simple, while sound effects depended on the functions available in KOPO. Another limitation was the lack of extended testing with actual kindergarten students. Without direct user testing, it is difficult to confirm that every child would understand the intended size order of the animals and vehicles. Immediate feedback was still retained because it helps learners notice mistakes and improve their next attempt (Hattie & Timperley, 2007; Shute, 2008).

3.3 Constraints and Design Trade-offs

Limited development time, KOPO functions, and differences in image quality created several **Constraints**. One major **Design Trade-off** involved choosing between visual variety and interface consistency. Three completely different layouts might have made each theme look more unique, but they could also have confused young learners. A shared template was therefore considered more suitable. Another decision involved balancing decoration with clarity. Colourful backgrounds, animation, and sound could improve **Engagement**, but too many visual or audio elements might distract students from the main ordering task. As a result, **Core Functionality** received higher **Priority**, especially clear instructions, visible objects, drop areas, feedback, and retrying. Multimedia elements were only kept when they supported the learning purpose (Mayer & Fiorella, 2021; Moreno & Mayer, 2007).

3.4 Lessons Learned and Action Plan

One important **Lesson Learned** was that instructions, image visibility, and platform limits should be checked before detailed visual work begins. Earlier testing could have identified the unclear fruit question and the limitations of KOPO animation and sound. In a future project, my **Action Plan** would begin with testing the platform functions, followed by creating a shared guide for image size, outlines, and backgrounds. Each prototype should also be checked against its Acceptance Criteria before being transferred into KOPO. Feedback from a teacher, parent, or someone familiar with kindergarten education would provide stronger evidence of **Acceptability**. Treating the Sprint Review as only a presentation would be an **Anti-pattern**, because useful feedback should lead to visible design changes. Recording problems and improvement actions during the **Sprint Retrospective** would also support **Continuous Improvement** in later work (Schwaber & Sutherland, 2020).

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